KO LIH HAN

Software Engineer | Applied AI & LLM Systems

+60 17-424 6282 | lihhanko526@gmail.com | github.com/kolihhan

EXPERIENCE

Kahap Inc | *Software Developer (Part-time)* Aug 2022 - Aug 2026

- Primary developer for a production feedback platform that centralized submissions, automated notifications, and admin workflows.
- Maintained a separate Laravel/Vue.js career platform, shipping features and resolving production application and database issues.
- Managed deployments and server operations, including migrations/upgrades, Laravel queues/workers, SSL renewal, and database troubleshooting.

Pac Com Solution | *Full-Stack Developer Intern* Jun 2021 - Jan 2022

- Developed and maintained Laravel/Vue.js features for production POS applications.

PROJECTS

Semantic Text-to-SQL | *Python · XiYanSQL 7B · LangGraph · FastAPI · SQLite*

- Built a local Text-to-SQL pipeline using XiYanSQL 7B with schema/value grounding, SQL validation, read-only execution, and bounded repair.
- Reached 58/100 execution-result matches on a 100-question BIRD evaluation and used failure analysis to refine grounding and repair.

Local RAG Search | *Python · BM25 · Qwen3-Embedding-0.6B · RRF · FastAPI*

- Built a hybrid retrieval API combining BM25, Qwen3 embeddings, and RRF, with per-result retrieval details for debugging.
- On 64 EnterpriseRAG-Bench queries over 5,189 documents, RRF improved Recall@10 from 0.734 to 0.763 and MRR@20 from 0.679 to 0.742 versus BM25.

M.S. Thesis - Workplace Implicit Intent | *Llama 3.1 8B · Retrieval · Knowledge Graphs · Semantic Evaluation*

- Built a role- and context-grounded evidence framework for listener-side workplace intent using 51,414 posts and 20,455 extracted utterances.
- On 401 held-out utterances, Role-Context Top1 raised automated semantic F1 from 0.6250 to 0.6890 over the no-retrieved-evidence baseline.

SKILLS

AI: RAG, Information Retrieval, LLM Evaluation, LangGraph

Programming / Backend: Python, SQL, JavaScript, PHP, FastAPI, REST APIs, Laravel, Vue.js

Data / Infra: MySQL, Redis, SQLite, AWS, Linux, Git, pytest

Languages: Chinese (Native), English (Fluent), Malay (Intermediate)

EDUCATION

National Tsing Hua University | *M.S., Information Systems & Applications* Sep 2024 - Jun 2026

GPA 4.09/4.3 | Taiwan

Chung Hua University | *B.S., Computer Science & Information Engineering* Sep 2022 - Jun 2024

GPA 3.9/4.0 | Taiwan